Government of Nepal
Ministry of Urban Development (MoUD)
Central Level Project Implementation Unit (CLPIU)
Babarmahal, Kathmandu

New RCC Type Designs and Other Designs Prepared by MoUD-CLPIU

6 June, 2017

Objectives

- To know about the designs carried out from CLPIU
  - RCC building non-compliance with NBC 205:1994
  - Dry Stone Masonry Building
- To know the information about the ongoing Designs at CLPIU

RCC Building Non-compliance with NBC 205:1994

11 numbers of designs have been carried out

1. 1 bay x 4 bay Building
2. 1 bay x 3 bay Building
3. 1 bay x 2 bay Building
4. Building having Cantilever projection 1.5 m
5. Building having Grid spacing 1.5 m c/c
6. Building with Four Number of Columns, all eccentric
7. Building having foundations at differential level

RCC Building Non-compliance with NBC 205:1994

11 numbers of designs have been carried out

8. Building having column size 9” x 12”
9. Building with a Missing Column
10. Building having Grid Size 4.5m x4.5m
11. Building having Grid size 3.5mx 5.5m

RCC Building Non-compliance with NBC 205:1994

- **Dead Load**

  (NBC 102:1994), (IS 875 (Part 1)-1987)

  Floor Finish: 1 KN/m2

- **Live Load**

  (NBC 103:1994), (IS 875 (Part 2) : 1987)

  All rooms and Kitchen: 2 KN/m2

  Toilet/ Bathroom: 2 KN/m2

  Corridor, Passage, Staircase, Balconies: 3 KN/m2

RCC Building Non-compliance with NBC 205:1994

- **Wall Load**

  Burnt Clay brick with cement mortar is used as infilled material.

  All external walls are made of **Full Brick Wall**

  Internal walls are made of **Half Brick Wall**

- **Soil Type**

  Soft soil type is considered for the analysis and design

RCC Building Non-compliance with NBC 205:1994

- **Fundamental Time Period**

  $T = 0.06\, H^{3/4}$

  Where, H= height of the Structure

- **Seismic Weight**

  Total Seismic Weight = Total Dead Load + 25% of Live Load

RCC Building Non-compliance with NBC 205:1994

- **Seismic Design Parameters**

Seismic Design has been carried out by using

NBC 105:1994

Basic Seismic Coefficient, ***C = 0.08***

Seismic Zoning factor, ***Z= 1***

Importance Factor, ***I= 1***

Structural Performance Factor, ***K = 1***

Design Horizontal Seismic Coefficient, $\boldsymbol{C_d}$***= CZIK= 0.8***

RCC Building Non-compliance with NBC 205:1994

- **Design Load Combination**

  1.5 (DL + LL)

  DL+1.3LL ± 1.25 EQ

  0.9 DL ± 1.25 EQ

  DL ± 1.25 EQ

  DL= Dead Load

  LL= Live Load

  EQ= Earthquake Load

RCC Building Non-compliance with NBC 205:1994

- **Method of Analysis**

Seismic Coefficient Method,

Horizontal Seismic Base Shear, $V = C_d W_t$

C_d *= Design Horizontal Seismic Coefficient*

W_t *= Seismic Weight of the Building*

Modal Response Spectrum Method

Modal response analysis is carried out considering more than 90% of the mass is participating in each direction

RCC Building Non-compliance with NBC 205:1994

- **Deflection Check**

The ratio of the inter-story deflection to the corresponding story height shall not exceed: 0.01

Nor shall the inter-story deflection exceed 60 mm.

RCC Building Non-compliance with NBC 205:1994

- **Detailing Code**

IS 13920:1993 is used for the detailing of the structural elements

RCC Building Non-compliance with NBC 205:1994

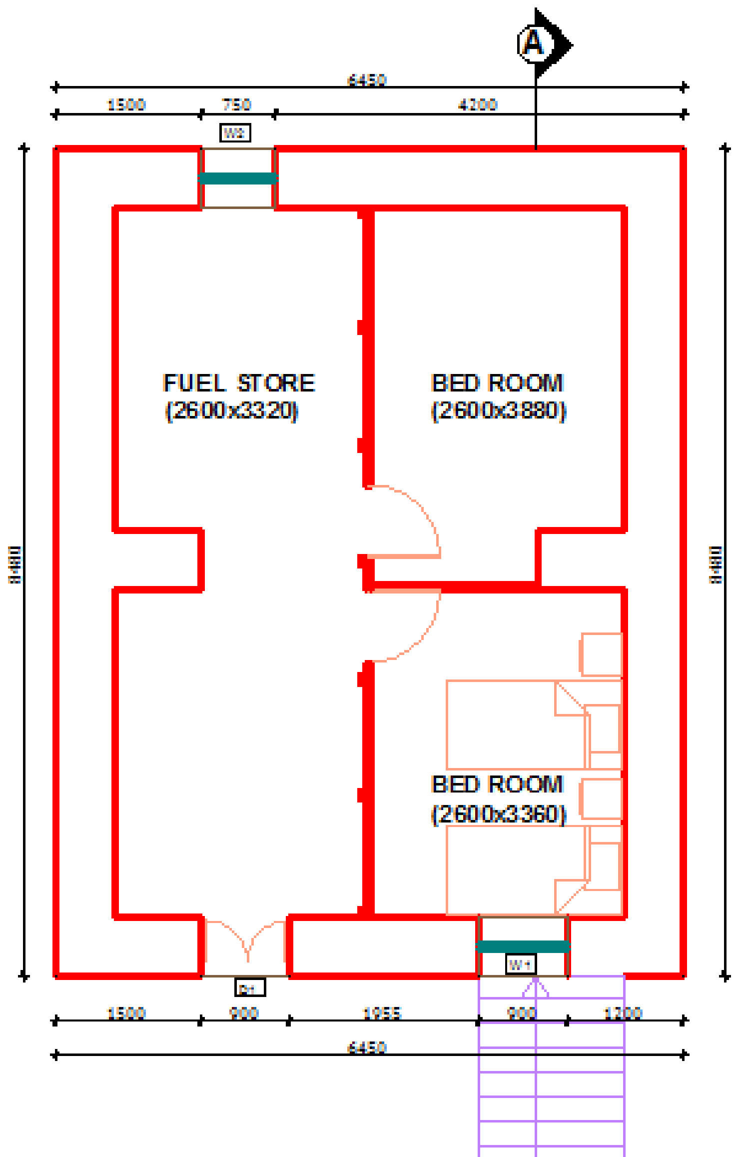
- **Construction of Structural Model**
  - Finite Element Based Structural Analysis software ETABS 2015 is used
  - Three dimensional model has been prepared.
  - Beam and Column members have been modeled using frame elements.
  - Shell elements have been used for modeling floor slab.

RCC Building Non-compliance with NBC 205:1994

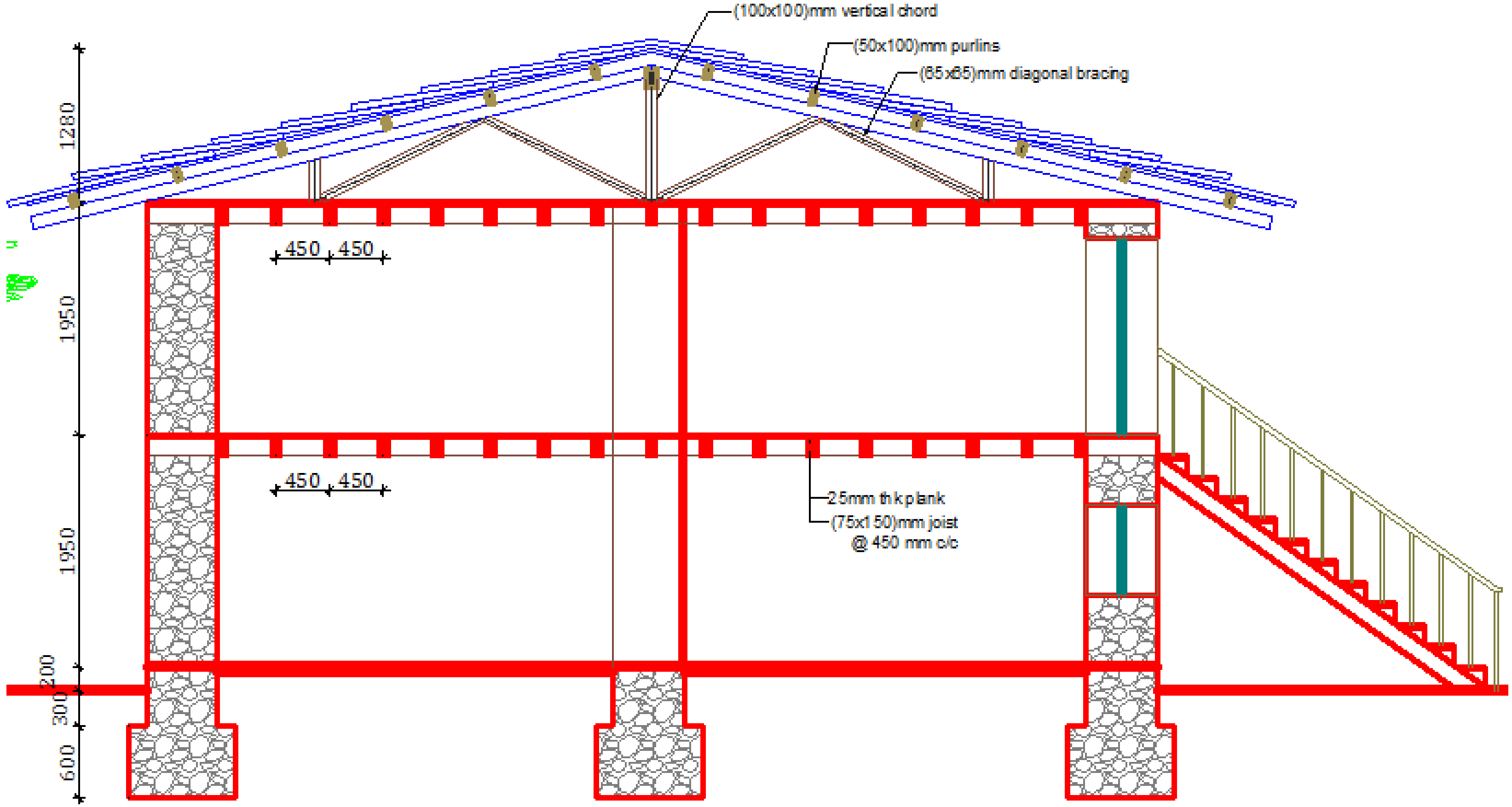
- **Construction of Structural Model**
  - The model is fixed at the basement level.
  - The columns and beams are modeled as rectangular sections.
  - Dead load and Live loads are applied as uniformly distributed loads per sq. meter on slab members.
  - Wall loads are applied as uniformly distributed loads per meter

Brief Description of Dry Stone Masonry

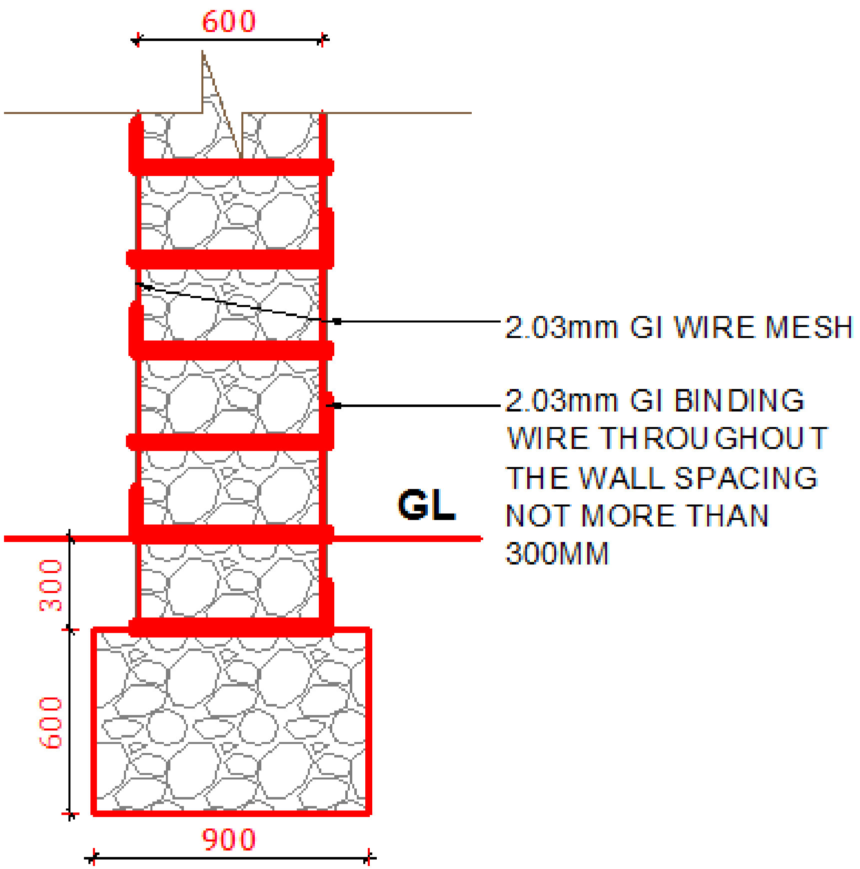
Dry Stone Masonry

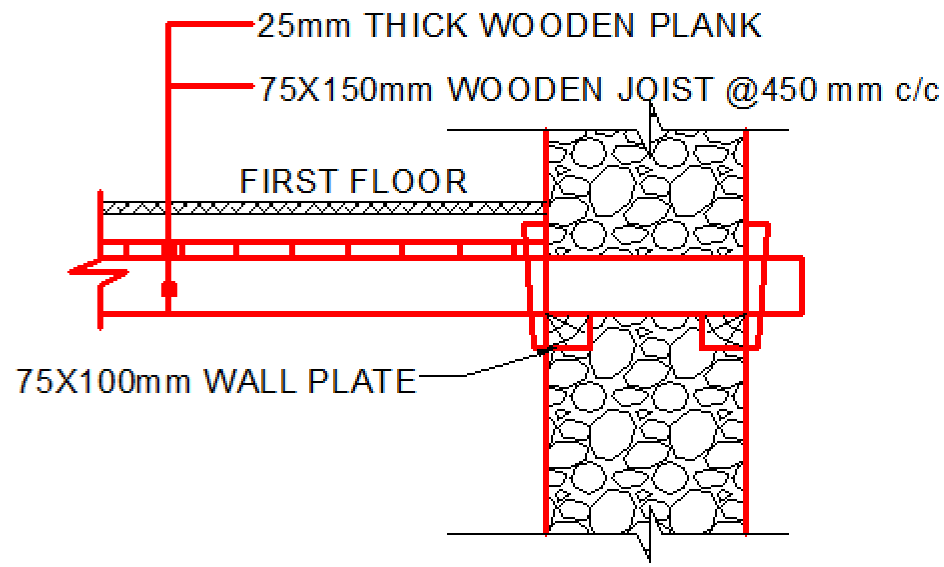


Dry Stone Masonry

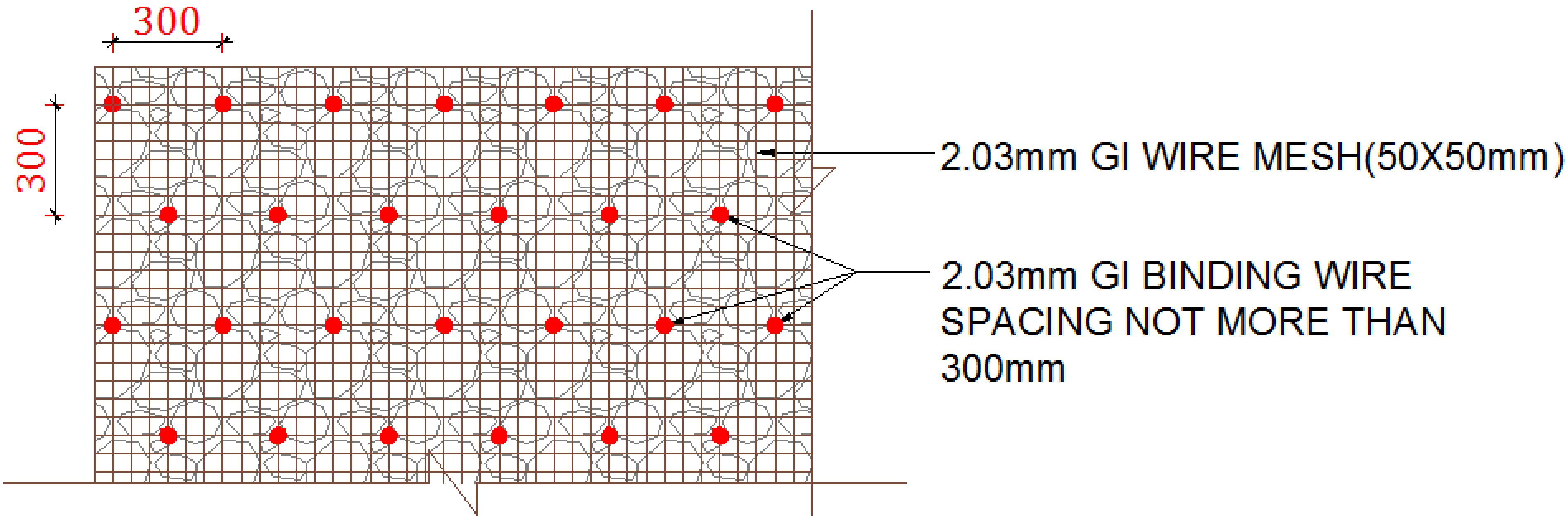


Dry Stone Masonry



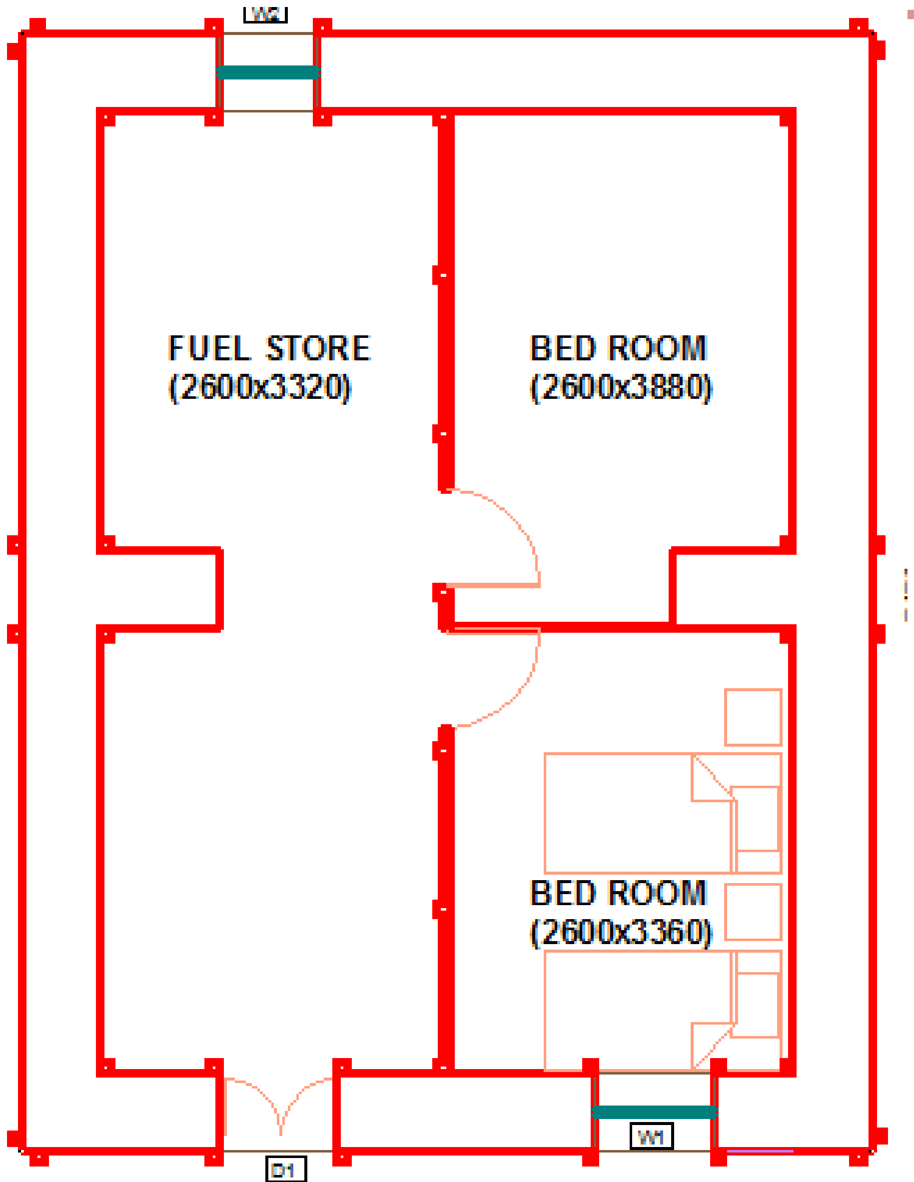


Dry Stone Masonry

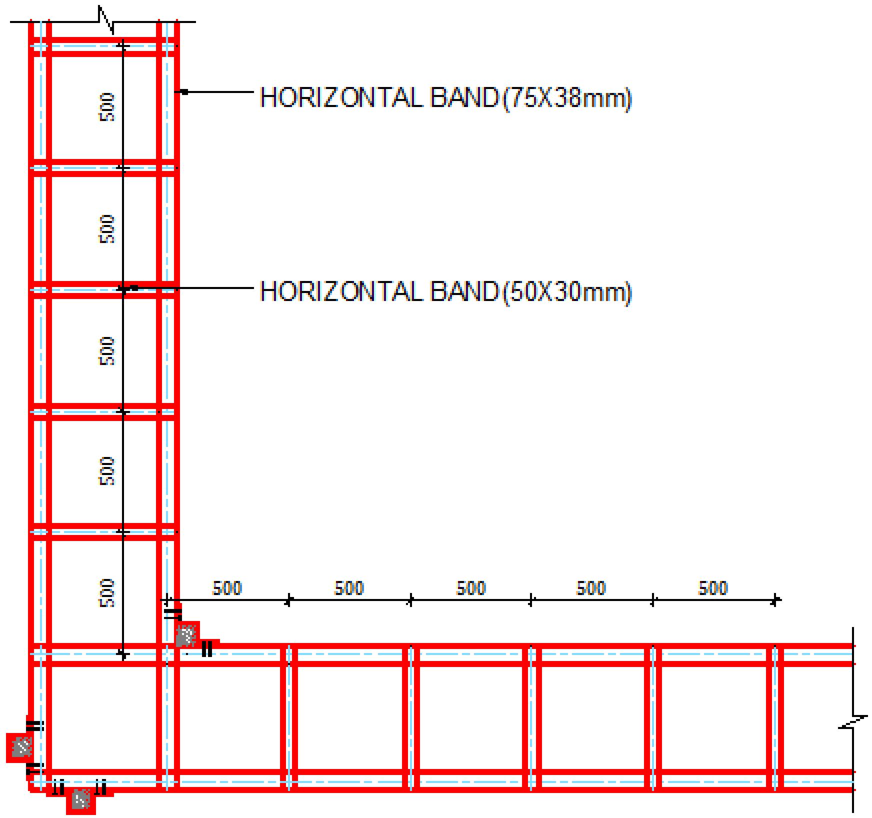


Dressed Dry Stone Masonry with Wooden Elements

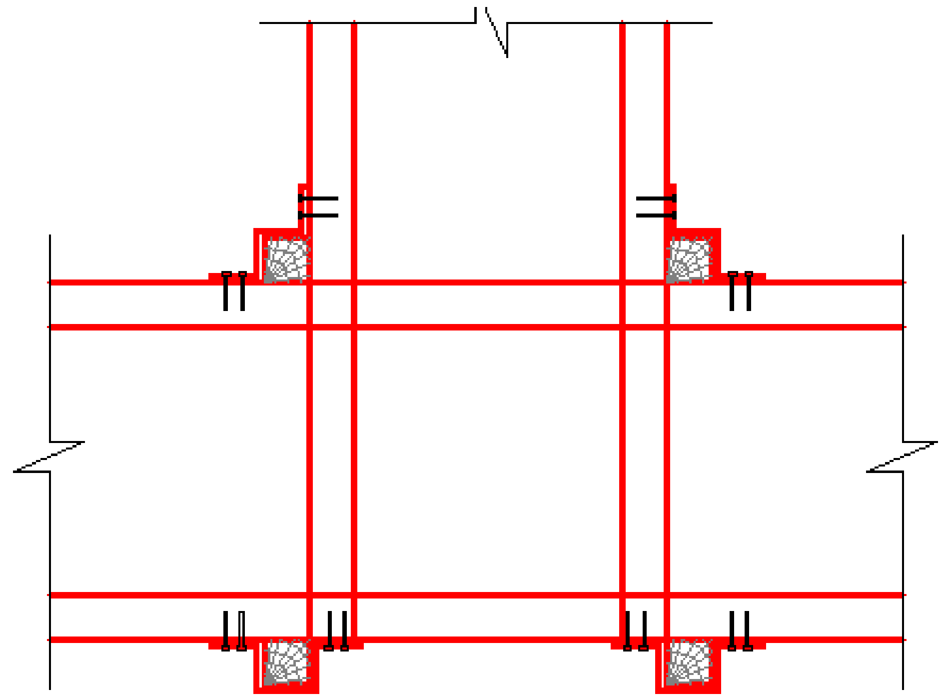
Dry Stone Masonry



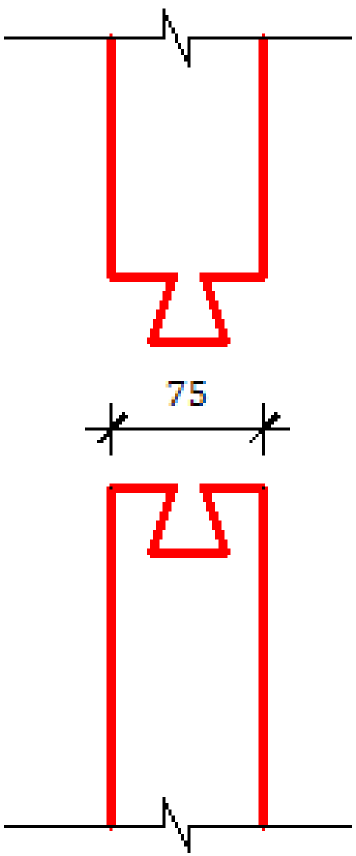
Dry Stone Masonry

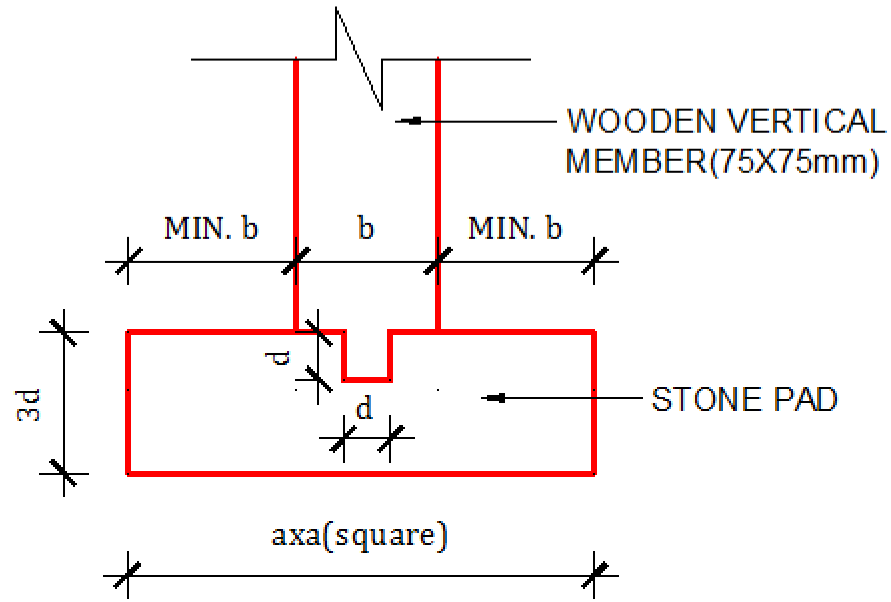


LINTEL SILL LEVEL WOODEN BAND ON LOAD BEARING WALLS



Dry Stone Masonry





DETAIL OF FIXING POST ON BASE PAD

a= 3b OR GREATER
b=SIZE OF POST
d=b/3 OR 50mm

Ongoing Designs

- Confined masonry building
- 9” x12” single storied Building
- Differential Level Foundation type 2: Two storied

Thank you